MIL-F-46220A (Ord)
20 April 1962
SUPERSEDING
MIL-F-46220 (Ord)
23 May 1960

MILITARY SPECIFICATION

FUZE, PD, M525A1E1,
HEAD ASSEMBLY FOR

1. SCOPE

1.1 This specification covers the Head Assembly only for Fuze, PD, M525A1E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Flat, Steel
UU-T-111 - Tape, Paper, Gummed (Sealing and Securing)
PPP-B-676 - Boxes, Fiber

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-117 - Bags, Exterior Packaging
MIL-B-2427 - Boxes, Ammunition Packing, Wood, Nailed
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-B-3106 - Board, Composition, Water Resistant, Solid (For Filler or Cushion Pads)
MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components
MIL-F-20416 - Fuze, PD, M52A2 and M52A2B1 and M519, Loading, Assembling and Packing
MIL-P-19644 - Plastic Foam Molded Polystyrene (Expanded Bead Type)
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

FSC: 1390

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STANDARDS

FEDERAL

Federal Test Standard 151

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-171 - Systems for Preparation, Painting and Finishing Metal and Wood Surfaces
MIL-STD-350 - Jolt Test for Use in Production of Fuzes.
MIL-STD-351 - Jumble Test for Use in Production of Fuzes.
MIL-STD-353 - Transporation Vibration Test for Use in Production of Fuzes.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

DRAWINGS

ORDNANCE CORPS

C8798660 - Plug, Slider.
C8798661 - Spring, Slider.
D8798662 or D8798663 - Body Assembly.
C8798677 - Slider.
D8799127 - Head Assembly.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-8815821 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

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2.2 Other publications.-The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials
ASTM Standards, 1958 Edition, Part 3.
(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material.-Material shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on drawing D8799127 and with all the requirements specified in applicable specifications, except as modified herein.

3.3 Head assembly.

3.3.1 Non-arming.-The assembly shall not arm when subjected to an acceleration of 150 plus 5 gravities (g's) applied in the direction simulating ballistic firing, when tested as specified in 4.4.1.

3.3.2 Transportation vibration.-The assembly shall withstand the test specified in Standard MIL-STD-353 when tested as specified in 4.4.2.

3.3.3 Non-Positioning.-The firing pin shall not move when a dead load of 10 plus 1 pound is applied to the striker assembly when tested as specified in 4.4.3 (see 6.3).

3.3.4 Positioning.-The firing pin shall move and shoulder on the slider when a dead load of 90 pounds, maximum, is applied to the striker assembly when tested as specified in 4.4.4.

3.3.5 Arming.-The assembly shall arm, as evidenced by firing pin withdrawal, after being subjected to an acceleration of 500-5's applied in the direction simulating ballistic firing, when tested as specified in 4.4.5.

3.3.6 Arming delay.-The assembly shall arm in not less than 1.8 or more than 3.5 seconds, when tested as specified in 4.4.6.

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3.3.7 Jolt and jumble.-The assembly shall withstand the tests specified in Standards MIL-STD-350 and MIL-STD-351, when tested as specified in 4.4.7.

3.4 Strength of train pinion-wheel teeth.-The teeth of the train gear and the teeth of the pinion shall withstand a torque of 40 inch-ounces, minimum, when tested as specified in 4.4.8.

3.5 Embrittlement of springs (firing pin spring and setback spring).-The springs shall not break when extended to the point of minimum permanent distortion when tested as specified in 4.4.9. (applicable to springs made of music wire only).

3.6 Assembly of firing pin and lower bushing assembly.-The load required to assemble the firing pin and lower bushing assembly shall not exceed 90 pounds, maximum.

3.7 Winding of head assembly.-The torque applied to the striker assembly during any winding or rewinding operation performed in the manufacture or testing of the head assembly shall not exceed 80 inch-ounces, maximum.

3.8 Porosity (applicable to die cast parts only).-There shall be no through porosity in any amount and no surface porosity in excess of 1/16 inch in any direction.

3.9 Workmanship.-All manufacturing, processing and assembly operations shall be correctly performed in a thoroughly workmanlike manner. Parts shall be of the proper finish (see Specification MIL-A-2550) and be free from burrs, chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements

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specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208 (see 6.4).

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. Unlisted defects will be subject to the provisions of MIL-A-2550.

4.2 Initial production inspection.

4.2.1 Submission.-The contractor shall submit an initial production sample consisting of 50 assemblies which have been produced by the process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. In addition, 25 complete sets of parts (i.e. 25 each of every individual component and every subassembly) shall be submitted, except that quantities for the following shall be:

175 train pinion-wheels
50 each of lower bushings, escape wheels, pallets, center gears, striker assemblies and upper plate assemblies
5 heads from each mold cavity
5 strikers from each mold cavity

The initial production sample shall be accompanied by certificates of analysis for all materials. All parts and materials shall be obtained from the same source of supply as will be used in regular production. An initial production sample shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, or specification, such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Each assembly and part shall be inspected by the Government for all the inspections specified in 4.3.2 and 4.3.3 of this specification and all requirements of the applicable drawings. The 50 head assemblies

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shall be subjected to the following tests in the order specified, except that only 12 of the assemblies shall be subjected to the jolt and jumble tests.

a. Transportation vibration (see 4.4.2)
b. Non-positioning (see 4.4.3)
c. Arming (see 4.4.5)
d. Arming delay (see 4.4.6)
e. Non-arming (see 4.4.1)
f. Positioning (see 4.4.4)
g. Jolt and jumble (see 4.4.7)

4.2.3 Rejection.-If any assembly or part fails to comply with the applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted or until otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of head assemblies produced by one manufacturer in one unchanged process (see 6.5) in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain components which have been produced in one unchanged process in accordance with the same drawing and drawing revision, same specification and specification revision, except that each of the following components shall be from not more than one manufacturer:

a. Center gear
b. Train pinion-wheel
c. Escape pinion-wheel
d. Lower bushing
e. Pallet

The lot identities applicable to components which are subject to the inspection requirements of material or process specifications (e.g., sintered components) shall be maintained throughout subsequent manufacture and processing. Data cards for head assembly lots shall be annotated as to these lot identities.

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4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Head (see dwg. C8799139 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Inspection Code No. (see 6.7)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Concentricity of movement assembly counterbore with pitch diameter of thread including perpendicularity to shoulder	Gage	01001
102.	Concentricity of striker cavity with movement assembly counterbore	Gage	01002
103.	Burr or rough surface on interior or in pull wire hole	Visual	01003
Minor:	AQL 4.00 percent		
201.	Diameter of striker cavity, maximum (max.)	Gage	01004
202.	Diameter of movement assembly counterbore, max	Gage	01005
203.	Pitch diameter of thread	Gage	01006
204.	Major diameter of thread	Gage	01007
205.	Length of thread, minimum (min.)	Gage	01008
206.	Diameter of shoulder, min	Gage	01009
207.	Porosity (see 3.8)	Visual	01010
208.	Evidence of poor workmanship (see 3.9)	Visual	01011

4.3.2.2 Striker assembly (see dwg. C8799135 covering a

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detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Concentricity of firing pin with outside diameter of striker	Gage	02001
102.	Surface finish of outside diameter improper	Visual	02002
Minor:	AQL 1.00 percent		
201.	Porosity (see 3.8)	Visual	02003
202.	Evidence of poor workmanship (see 3.9)	Visual	02004

4.3.2.3 Spring, firing pin (see dwg. C8799128 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Load test failure, max	Gage	03001
102.	Load test failure, min	Gage	03002
103.	Outside diameter	Gage	03003
Minor:			
201.	Evidence of poor workmanship (see 3.9)	Visual	03004

4.3.2.4 Lower plate assembly (see dwg. B8799156 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Hardness of plate, min	Test (see 4.4.12)	04001

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102.	Location of posts in relation to setback slot and center hole...	Gage	04002
103.	Width of setback slot, min.....	Gage	04003
104.	Height of setback lug.....	Gage	04004
105.	Width of slot on center hole, min.	Gage	04005
106.	Diameter of post hole extrusions, max.....	Gage	04006
107.	Thickness of plate, max.....	Gage	04007
108.	Burr.....	Visual	04008
109.	Post missing or loose.....	Visual-Manual	04009

Minor: AQL 0.40 percent

201.	Evidence of poor workmanship (see 3.9).....	Visual	04010

4.3.2.5 Upper plate assembly (see dwg. C8799149 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major: AQL 0.65 percent

101.	Width of setback slot, min.....	Gage	05001
102.	Location of blind holes in relation to setback slot and center hole.....	Gage	05002
103.	Center gear tooth missing, cracked, chipped or malformed.....	Visual	05003
104.	Washer cracked.....	Visual	05004
105.	Burr.....	Visual	05005
106.	Center gear does not rotate freely	Manual	05006

Minor: AQL 0.40 percent

201.	Evidence of poor workmanship (see 3.9).....	Visual	05007

4.3.2.6 Setback (see dwg. C8800955 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:		AQL 0.40 percent	
101.	Profile (after plating).....	Gage	06001
102.	Thickness.....	Gage	06002
103.	Flatness.....	Gage	06003
104.	Burr.....	Visual	06004
Minor:		AQL 1.00 percent	
201.	Evidence of poor workmanship (see 3.9).....	Visual	06005
202.	Protective finish inadequate.....	Visual	06006

4.3.2.7 Pinion wheel, escape (see dwg. C8799143 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:		AQL 1.00 percent	
101.	Concentricity of diameter over escape teeth with center hole.....	Gage	07001
102.	Profile of escape teeth, max.....	Gage	07002
103.	Total height, min.....	Gage	07003
104.	Thickness of escape teeth, max....	Gage	07004
105.	Diameter of center hole, min.....	Gage	07005
106.	Excessive flash (must not enter area bounded by angle on bottom of component).....	Visual	07006
107.	Escape or pinion tooth missing, cracked, chipped or malformed.....	Visual	07007
108.	Angle on bottom missing or inadequate.....	Visual	07008
Minor:		AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.9).....	Visual	07009

4.3.2.8 Pinion wheel, train (see dwg. C8799141 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:		AQL 0.65 percent	
101.	Concentricity of center hole with pitch diameter of pinion	Gage	08001
102.	Concentricity of center hole with pitch diameter of train gear	Gage	08002
103.	Diameter of center hole, min	Gage	08003
104.	Train gear or pinion tooth missing, cracked, chipped or malformed	Visual	08004
105.	Excessive flash (must not enter into area bounded by angle on bottom of component)	Visual	08005
106.	Angle on bottom missing or inadequate	Visual	08006
Minor:		AQL 0.40 percent	
201.	Evidence of poor workmanship (see 3.9)	Visual	08007

4.3.2.9 Center gear (see dwg. C8799151 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Concentricity of hub diameter with pitch diameter of gear teeth	Gage	09001
102.	Concentricity of hub diameter with diameter of hex hole	Gage	09002
103.	Radius on hex hole missing	Visual	09003
104.	Burr in hex hole	Visual	09004
105.	Excessive flash (must not enter area bounded by angle on bottom of gear)	Visual	09005
106.	Tooth missing, cracked, chipped malformed	Visual	09006
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	09007

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4.3.2.10 Pallet (see dwg. C8799142 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Profile	Gage	10001
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	10002

4.3.2.11 Bushing, lower (see dwg. C8799154 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None definded			
Major:	AQL 0.40 percent		
101.	Concentricity of hex hole with outside diameter	Gage	11001
102.	Width of slot between top of external lugs and flange	Gage	11002
103.	Burr	Visual	11003
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	11004

4.3.2.12 Bushing, spring (see dwg. C8799140 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Thickness of flange	Gage	12001

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Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.9).....	Visual	12001

4.3.2.13 Wire, pull (see dwg. C8799130 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.5 percent		
201.	Diameter across coil.....	Gage	13001
202.	Chamfer missing on end of straight portion.....	Visual	13002
203.	Burr on end of straight portion.	Visual	13003
204.	Evidence of poor workmanship (see 3.9).....	Visual	13004

4.3.2.14 Movement assembly, prior to assembly of spring bushing (see dwg. C8799138 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Distance between upper and lower plates under load.....	Gage	14001
102.	Any component missing, damaged or improperly assembled.....	Visual	14002
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.9).....	Visual	14003

4.3.2.15 Movement assembly (see dwg. C8799138 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.25 percent		
101.	Movement fails to operate properly	Manual (see 4.4.13)	15001

Minor: None defined

4.3.2.16 Pull wire assembly (see dwg. C879910006 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.00 percent		
201.	Length of cord between clips	Gage	16001
202.	Clip missing or insecure (suspect assemblies should be subjected to the load specified on the drawing)	Visual-Manual	16002
203.	Evidence of poor workmanship (see 3.9)	Visual	16003

4.3.2.17 Pin, firing (see dwg. B8799137 covering a detail of dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Depth, pitch and angle of serrations	Gage	17001
102.	Profile of point	Gage	17002
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.9)	Visual	17003

4.3.2.18 Other components.-All head assembly component parts and subassemblies not specifically listed in 4.3.2.1 through 4.3.2.17 above shall be subjected to the following inspections (see 3.9 and applicable drawings)

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Operation missing or improperly performed	Visual	18001
102.	Component missing or improperly assembled	Visual	18002
Minor:	AQL 0.40 percent		
201.	Protective finish inadequate or other evidence of poor workmanship (see 3.9)	Visual	18003

4.3.2.19 Head assembly (see dwg. D8799127)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Distance from end of lower bushing to flange, min, or firing pin in armed position	Gage-Visual	19001
2.	Pull wire missing, insecure, or not engaged in setback	Visual-Manual	19002
Major:	AQL 1.00 percent		
101.	Distance from shoulder to setback, max	Gage	19003
102.	Distance from shoulder to firing pin point	Gage	19004
103.	Firing pin misaligned in hex hole	Visual	19005
104.	Firing pin point damaged	Visual	19006
105.	Movement assembly loosely crimped or not crimped full 360 degrees (no "play" permitted in any direction)	Manual	19007
106.	Lower bushing cracked	Visual	19008
107.	Lower plate deformed by crimping operation as evidenced by broken or cracked extrusions on post holes, buckling or bending	Visual	19009
108.	Cord assembly missing, insecure		

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	or not properly assembled	Visual	19010
Minor:	AQL 1.50 percent		
201.	Threads damaged	Visual	19011
202.	Evidence of poor workmanship (see 3.9)	Visual	19012
203.	Marking misleading or unidentifiable	Visual	19013

4.3.2.20 Pad, polystyrene (see 5.1.1)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Thickness, min. (to be measured at center of pad as well as near edge)	Gage	20001
102.	Evidence of poor workmanship (see 3.9)	Visual	20002
Minor:	AQL 1.00 percent		
201.	Diameter of holes	Gage	20003
202.	Evidence of poor workmanship (see 3.9)	Visual	20004

4.3.2.21 Sealed carton (see 5.1.1)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bag improperly sealed	Visual	21001
102.	Bag pierced or torn	Visual	21002
103.	Contents loose	Manual	21003
Minor:	None defined		

4.3.2.22 Sealed wood packing box (see 5.2.1)

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:		AQL 0.25 percent	
101.	Box damaged	Visual	22001
102.	Strapping missing, broken or loose	Visual	22002
Minor:		AQL 1.50 percent	
201.	Marking misleading or unidentifiable	Visual	22003
202.	Strapping improperly assembled	Visual	22004
203.	Nail protruding	Visual	22005
204.	Contents loose	Visual	22006

4.3.3 Testing.

4.3.3.1 Non-arming test (see 3.3.1) Critical defect.-This test shall be performed 100 percent. Any assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1 (Code Number 19014)

4.3.3.2 Transportation vibration (see 3.3.2) Critical or Major defect, as noted herein.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter L and AQL 0.25 percent, except that no critical defects shall be permitted and a determination regarding acceptance shall not be made until the non-positioning and positioning tests have been conducted (see 4.3.3.3 and 4.3.3.4). If any assembly is not safe after the test (i.e., it has become armed or partially armed), it shall be considered a critical defect. If any assembly is not operable after the test, it shall be considered a major defect. The test shall be performed as specified in 4.4.2. (Code Number 19015)

4.3.3.3 Non-positioning test (see 3.3.3) Major defect.-The same assemblies shall be used for this test as were used in the transportation vibration test. Acceptance or rejection shall be determined on the results of the two tests combined. The test shall be performed as specified in 4.4.3. (Code Number 19016)

4.3.3.4 Positioning test (see 3.3.4) Critical defect.-The same assemblies shall be used for this test as were used in the transportation vibration and non-positioning tests. If any sample fails to comply with the applicable requirement the lot shall be rejected. The test shall be performed as specified in 4.4.4 (Code Number 19017)

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4.3.3.5 Arming test (see 3.3.5) Major defect.-Sampling plans and procedures for this test shall be in accordance with Standard MIL-STD-105 and AQL 0.25 percent. Any assembly in which the firing pin does not retract after having been subjected to the arming acceleration, regardless of cause, shall be classed defective. The test shall be performed as specified in 4.4.5. (Code Number 19018)

4.3.3.6 Arming delay test (see 3.3.6) Minimum delay, Critical defect; Maximum delay, Major defect.-Sampling plans and procedures for this test shall be in accordance with Standard MIL-STD-105 and AQL 0.25 percent except that the sample size shall never be less than 300. No critical defects shall be permitted and any assembly which does not complete the timing cycle in the required time, regardless of cause, shall be classed defective. The test shall be performed as specified in 4.4.6. (Code Number 19019)

4.3.3.7 Jolt and jumble tests (see 3.3.6 and 3.3.7) Critical defect.-A sample of 12 head assemblies shall be selected for these tests. If one or more assemblies fail to comply with any of the applicable requirements the lot shall be rejected. Testing shall be performed as specified in 4.4.7 (Code Numbers: Jolt, 19020; Jumble, 19021)

4.3.3.8 Torque tests of train pinion-wheel(see 3.4 and dwg. C8799141) Major defect.-Sampling plans and procedures for each torque test shall be in accordance with Standard MIL-STD-414, Section B or C, Table B1 or C1, Code letter C and AQL 0.65 percent. In addition, the lot shall be rejected if two or more teeth in any sample quantity fail to meet the applicable minimum requirement. The test shall be performed as specified in 4.4.8. The same train pinion-wheels used in the testing of train gears shall not be used in the testing of pinions; separate sample units shall be selected. (Code Numbers: Train gear; 08008; Pinion, 08009)

4.3.3.9 Embrittlement of springs (see 3.5 and dwgs. C8799128 and B8800956) Major defect.-A sample of 30 springs each (spring, firing pin, and spring, setback) from each lot shall be selected for testing in accordance with 4.4.9. If one or more springs fail to meet the requirements specified the lot shall be rejected. (Code Numbers: Firing pin spring, 03004; Setback spring, 23001)

4.3.3.10 Assembly of firing pin and lower bushing (see 3.6 and dwg. D8799127).-The contractor shall maintain a positive process control which will prevent the assembly of any head

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assembly in which the firing pin cannot be inserted into the lower bushing assembly with a maximum load of 90 pounds.

4.3.3.11 Winding of head assembly (see 3.7 and dwg. D 8799127).-The contractor shall maintain a positive process control during all winding operations which will prevent the application of torque to the striker which is in excess of the applicable requirement. (This requirement is interpreted to apply to all rewinding operations as well as initial winding.)

4.3.3.12 Pull test of striker assembly (see dwg. C8799135) Major defect.-Sampling plans and procedures for this test shall be in accordance with Standard MIL-STD-105, Code letter L and AQL 0.25 percent. The sample quantity shall be chosen so as to include strikers from all cavities in the striker die(s) in approximately equal proportion. The test shall be performed as specified in 4.4.10. (Code Number 02005)

4.3.3.13 Load test of upper plate assembly (see dwg. B8799149) Major defect.-A sample of 50 assemblies shall be selected for this test. If two or more assemblies fail to comply with the applicable requirement, the lot shall be rejected. If only one defective is found, a second sample of 100 assemblies shall be selected. If the combined number of defectives in both the first and second samples is two or more, the lot shall be rejected. The test shall be performed as specified in 4.4.11. (Code Number 05008)

4.3.3.14 Lower bushing (see dwg. C8799154) Major defect.-A sample of 30 lower bushings shall be selected from each lot. Each sample unit shall be sectioned at right angles to the internal slots and be examined for compliance with the requirement for the maximum radius at the bottom of the internal slots and for any foreign matter which would interfere with proper seating of the tubular clip clutch. If one or more bushings fail to comply with the applicable requirement, the lot shall be rejected. (Code Number 11005)

4.3.3.15 Plating of sintered components (see dwg. C8799141, C8799142, C8799143, C8799151 and C8799154).-For each component, determination of compliance with the applicable requirements for plating shall be in accordance with the provisions for Finish No. 1.1.2.2 of Standard MIL-STD-171, except that sampling and acceptance criteria as applicable shall be as modified herein. In all tests actual components shall be used (separate panels or specimens are not permitted).

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4.3.3.15.1 Salt spray test, Major defect.-A random sample of 10 units shall be selected from each inspection lot of each component for this test. Failure of one or more sample units to meet the applicable requirement shall be cause for rejection (see 6.6) (Code Numbers: Escape wheel, 07010; Train pinion-wheel, 08010; Center gear, 09008; Pallet, 10003; Lower bushing, 11006)

4.3.3.15.2 Plating adhesion test, Major defect.A random sample of 10 units shall be selected from each inspection lot of each component for this test. Failure of one or more sample units to comply with the applicable requirement shall be cause for rejection. (Code Numbers: Escape wheel, 07011; Train pinion-wheel, 08011; Center gear, 09009; Pallet, 10004; Lower bushing, 11007)

4.3.3.15.3 Plating thickness test (microscopic examination), Major defect.-A random sample of 10 units shall be selected from each inspection lot of each component for this test. A minimum of four readings shall be taken on each unit and an average obtained. If any individual reading is less than .00025 inch or if the average reading of any unit is less than .00030 inch, the lot shall be rejected. (Code Numbers: Escape wheel, 07012; Train pinion-wheel, 08012; Center gear, 09010; Pallet, 10005; Lower bushing, 11008)

4.3.3.16 Density of sintered components (see dwgs. C8799141, C8799142, C8799143, C8799151, C8799154) Major defect.-A sample of 10 units shall be selected from each inspection lot of each component for this test (after sintering and prior to plating). If any sample unit fails to comply with the applicable requirement the lot shall be rejected. The test shall be performed as specified in 4.4.14. (Code Numbers: Escape wheel, 07013; Train pinion-wheel, 08013; Center gear, 09011; Pallet, 10006; Lower bushing, 11009)

4.3.3.17 Leakage test of sealed carton (see 5.1.1.1) Major defect.-Sampling, acceptance criteria and test procedures for this test shall be as prescribed in specification MIL-P-116 for the "Quick Leak Test". (Code Number 21004)

4.3.4 Inspection equipment.-Index of Inspection Equipment Lists Number 8815821 identifies available approved inspection equipment designs. The contractor may, however, design alternate inspection equipment in order to facilitate his operation. Designs of any acceptance inspection equipment prepared by the contractor to support his approved written inspection plan (see 4.1.1) shall be approved by the Government prior to fabrication or procuring of such equipment (see 6.2). All acceptance inspection equipment

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shall be certified by the Government as complying with the approved design prior to its use. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Non-arming.-The unarmed head assemblies, less pull wire, shall be mounted in a centrifuge with the axis of the firing pin parallel with the rotating arm, and the pointed end of the firing pin away from the axis of rotation. The g's acceleration shall be measured to the center of gravity of the setback.

4.4.2 Transportation vibration.-The unarmed head assemblies, with pull wire, shall be tested in accordance with the procedure specified in Standard MIL-STD-353 with the exception that any disassembly shall not be made until other tests specified in 4.2 or 4.3.3.3 and 4.3.3.4 have been completed, as applicable. Operation shall be determined by manually or mechanically releasing the setback and observing for proper functioning of the unit, i.e., retraction of the firing pin after normal completion of the timing cycle.

4.4.3 Non-positioning.-The head assemblies shall be placed in a fixture which supports the lower bushing and the specified load applied to the striker.

4.4.4 Positioning.-The unarmed head assemblies shall be assembled to fuze body assemblies conforming to dwg. D8798662 or D8798663 complete with empty (unloaded) sliders conforming to dwg. C8798677 and the specified load applied to the striker. Observation of the interior of the slider cavity shall be made with an otoscope (or other approved instrument) in order to determine that the firing pin has actually dented the slider and that the firing pin shoulder has positioned against the flat surface of the slider.

4.4.5 Arming.-The unarmed head assemblies, less pull wire, shall be tested as specified in 4.4.1 using the arming acceleration speed.

4.4.6 Arming delay.-The setback of the unarmed assemblies, less pull wire, shall be manually or mechanically initiated. The interval between initiation of the setback and firing pin arming shall be measured with a timing device to the nearest 0.1 second.

4.4.7 Jolt and jumble.-The sample unarmed head assemblies,

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with pull wire assembled, shall be assembled to fuze body assemblies, dwg. D8798662 or D8798663, complete with empty sliders, dwg. C8798677, slider plugs, dwg. C8798660 and slider springs, dwg. C8798661. The assemblies shall be positioned by applying the required load and then be tested in accordance with the procedures specified in Standards MIL-STD-350 and MIL-STD-351 except that the jumble test shall be conducted utilizing a jumble fixture as specified in Specification MIL-F-20416. An assembly shall be considered to have successfully passed the tests if, after disassembly of the slider plug and spring, the slider is retained in its cavity by the detented firing pin as determined by observation with an otoscope (or other approved instrument).

4.4.8 Torque test of train pinion-wheel.-The train pinion-wheel shall be placed in a fixture and be secured in position by a device that will lock against the teeth of the gear as to prevent it from moving. A lever shall contact one tooth tangentially at the calculated pitch circle. Torque shall be applied to the lever in the direction which will maintain the point of contact and so that the torque applied to the tooth being tested is at the correct pressure angle. Torque shall be applied slowly and evenly without impact until the tooth fractures. The maximum force required shall be recorded to the nearest inch-ounce. The internal friction of the test equipment shall not exceed one inch-ounce.

4.4.9 Embrittlement of springs.-The representative springs shall be extended to the minimum point of permanent distortion by any method satisfactory to the Government inspector.

4.4.10 Pull test of striker assembly.-The striker assembly shall be retained in a suitable fixture while a pulling force is exerted on the firing pin at a rate of 0.5 inch per minute, max. until a force of 50 pounds has been applied.

4.4.11 Load test of upper plate assembly.-The upper plate assembly will be positioned in a fixture designed to support the plate and to allow movement of the center gear. A dead load of 100 pounds shall be applied to the crimped side of the assembly.

4.4.12 Hardness test of lower plate.-This test shall be performed as specified in Federal Test Standard 151, Method No. 243.

4.4.13 Operation of movement assembly.-The movement shall be

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held in a vertical position, lower plate down, and a striker assembly (or other appropriate instrument) inserted into the center gear and rotated for 360 degrees in a clockwise direction. Then the movement shall be inverted and this inspection repeated by again rotating the striker assembly clockwise for 360 degrees, thus reversing the direction of the gear train operation. Each 360 degrees rotation shall be accomplished evenly, without impact, in not less than 10 seconds. Any movement which does not run freely and smoothly (i.e., it jams, catches, binds or locks), or is found to contain a broken gear, or in which the gears become disengaged, shall be rejected.

4.4.14 Density of sintered components.-Density shall be determined in accordance with Method B328-58T, ASTM standards, 1958 Edition, Part 3. Each sample unit shall be weighed three times and the average of the three weights used for the actual density calculation.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-Packaging for head assemblies for Fuzes, PD, M525A1E1 shall be as described below.

5.1.1.1 Head assembly.-Two hundred and fifty head assemblies shall be packed in five layers in a Style RSC, "B" Flute 275 pounds minimum test, corrugated fiberboard, Specification PPP-B-636, carton. Inside dimensions of the carton shall be 14 15/16 by 7 7/16 by 11 1/2 inches. Each carton requires ten pads constructed of molded polystyrene, Type N, 2 pounds to 2 1/4 pounds per cubic foot density, Specification MIL-P-19644. Each pad shall have fifty 27/32 inch diameter holes arranged in a 10 by 5 pattern. The holes shall be located 11/16 inch from the edges and be 1 1/2 inches between centers. Over-all dimensions of the pad shall be 14 7/8 by 7 3/8 by 0.6 inch thick, minimum. Each carton requires five egg-crate separators, "B" Flute, 275 pounds minimum test, corrugated fiberboard, Specification PPP-B-636. Each separator shall have partitions arranged in a 10 by 5 pattern and shall be 1 3/8 inches square. The height of the separators shall be 1 1/16 - 1/16 inch. A spacer, 275 pounds minimum test, "B" Flute, corrugated fiberboard, Specification PPP-B-636, 7 3/8 inches square, shall be placed in the void between the inner

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flaps (top and bottom). Place one pad on bottom of carton. Place a separator on top of the pad. Pack 50 head assemblies, firing pin downward, into the compartments of the separator with firing pins resting inside pad holes. Place two pads on top of head assemblies, so that the top portion of head assemblies rests inside the holes of the lower pad. Place a separator on top and pack 4 more layers in the same manner. One pad shall be placed on the top layer of head assemblies. Add suitable filler material on top of contents to insure tight packing. Seal carton in accordance with Specification MIL-P-10025 and blunt carton corners. Place carton in a Type 1, Class E, barrier material, Specification MIL-B-117, bag. The inside dimensions of the bag shall be 20 1/2 inches minimum by 26 inches minimum. Exhaust excess air in bag and seal in accordance with manufacturer's instructions with seal located at edge of bag to provide for a minimum of two subsequent seals, if required. After sealing the barrier bag, the package shall comply with the requirements for the "quick leak test" of Specification MIL-P-116.

5.2 Packing.

5.2.1 Level A.-Packing for head assemblies for Fuzes, PD, M525A1E1 shall be as described below.

5.2.1.1 Head assembly.-Two cartons (500 head assemblies) shall be packed in a Grade A, Type 1, Class 1, Specification MIL-B-2427 box. The inside dimensions shall be 15 3/4 by 15 11/16 by 12 5/16 inches. Place one end filler, 15 1/2 by 12 by 1/16 inch thick, Type I or II composition board, Specification MIL-B-3106 along each end of box. Place one top filler, 15 1/2 by 15 1/2 by 1/16 inch thick, same material, on top of contents. Add more fillers of the same material along sides, top and ends as required to insure tight packing. Secure box with two straps, steel, Type I, Class A or B, Specification QQ-S-781. Strap to be 5/8 by .02 by 68. inches long and shall be applied circumferentially inside the cover cleats.

5.3 Marking.

5.3.1 Level A.-Box and inner package marking shall comply with Standard MIL-STD-129. Alternatively, for domestic shipment, boxes and inner packages may be marked in accordance with Specification MIL-P-10025.

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5.4 Marking.-Marking of all packaging and packing shall comply with Standard MIL-STD-129.

5.5 Data cards.-Data cards shall be prepared for each lot in accordance with the provisions in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Submission of contractor developed inspection equipment designs.-Documentation of contractor developed inspection equipment designs shall be submitted in duplicate to Picatinny Arsenal, Dover, New Jersey, Attention ORDBB-ND, for approval. This documentation shall consist of a detailed technical description of each design, supported by drawings or sketches when necessary, and a specific statement concerning the use and application of the equipment including reference to the associated examination or test required by the specification. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.3 Movement of the firing pin.-Movement is defined as slippage of the firing pin so that the clutch engages in the next or subsequent serration from its original location.

6.4 Quality assurance provisions of other documents.-The quality assurance provisions of other documents which are required to be a part of the contractor's written description include, but are not limited to, those which relate to plating or other protective coating and to physical, chemical and other properties, for which specific quality assurance of inspection provisions are included in the applicable material and process specifications.

6.5 Change in process.-In addition to production operations the process includes all inspection operations - screening, sampling, etc. which are conducted by the contractor. A change in any contractor production or inspection operation is construed as a change in process.

6.6 Salt spray test.-At the discretion of the inspector, a lot may be provisionally accepted pending the results of the salt spray test. However, attention is directed to the provisions of

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4.3.1, which assure that in the event of test failure, the rejected material can later be identified.

6.7 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate data collection and analysis by the Government.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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